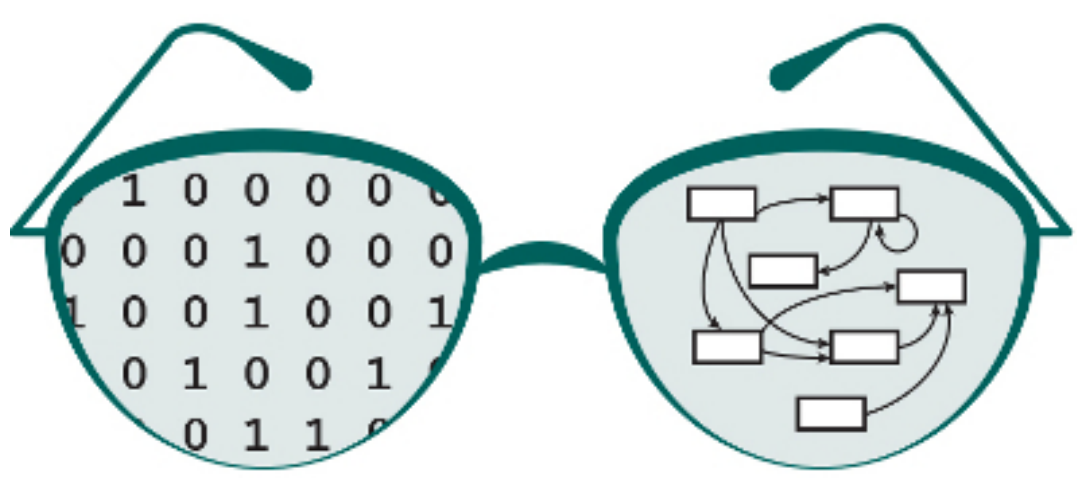




# Improved User Interface and Export Functionality for GRNsight 7.6, a Web Application for Visualizing Gene Regulatory and Protein-Protein Interaction Network Models

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<http://dondi.github.io/GRNsight/>



Loyola Marymount University

GRNsight, automatically lays out a gene regulatory network (GRN) graph, coloring the edges based on regulatory weights, and the nodes based on experimental or simulated expression data

## 1. Network

- Demo files are provided.
- Can import and export Excel, SIF, or GraphML files.
- Can construct a network from SGD regulation or PPI data stored in a backend database.

## 2. Layout

- Users can select a force graph layout or grid layout.
- Force graph parameter sliders:
  - Link distance determines the minimum distance between nodes.
  - Nodes have a charge which repels or attracts other nodes.

## 3. Node

- Coloring can be based on user-uploaded expression data or from published expression datasets stored in a backend database.
- Datasets available:
  - Apweiler et al. 2012, GSE33098
  - Barreto et al. 2012, GSE24712
  - Dahlquist et al. 2018, GSE83656
  - Kitagawa et al. 2002, GSE9336
  - Thorsen et al. 2007, GSE6068

## 4. Edge

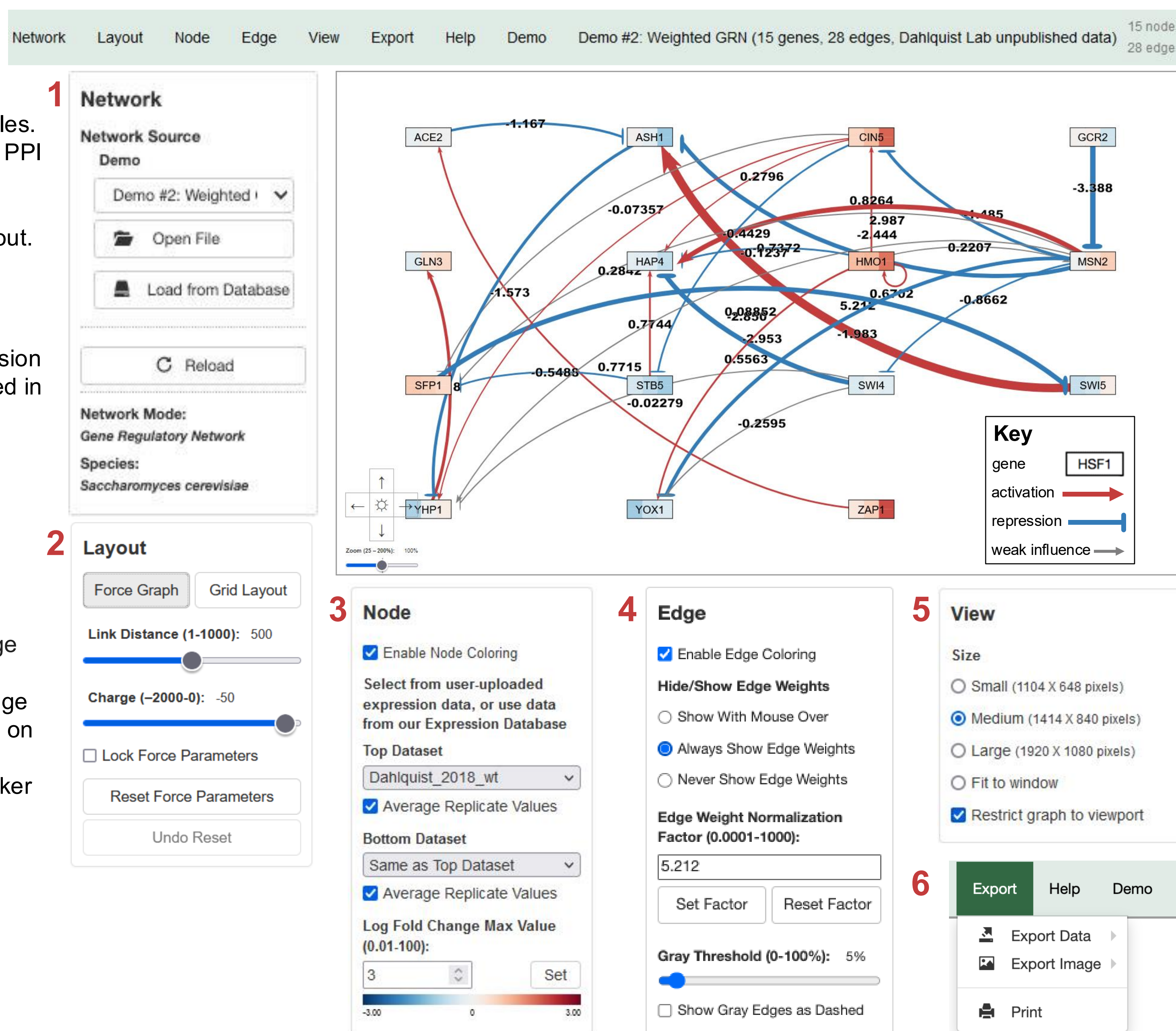
- Edge coloring can be toggled on and off.
- Buttons enable the user to always/never see edge weights or see them upon mouseover.
- Allows user to set normalization factor so that edge thicknesses for different graphs can be rendered on the same scale.
- Changing the gray threshold deemphasizes weaker edges in the visualization.
- Presenting gray edges as dashed lines further deemphasizes weak edges.

## 5. View

- Multiple viewport sizes available.
- Graph bounding box can be separated from viewport.
- Zooming and scrolling available.

## 6. Export

- Basic data export to SIF or GraphML or full GRNmap-compatible Excel workbook, with yeast mRNA degradation rates from Neymotin et al. (2014).
- Image export to PNG, SVG, or PDF.



Nodes can be colored with mRNA-level expression data imported by the user or from the backend expression database

Node coloring can also be toggled on or off. GRNsight needs to take this switch into account when exporting data to an Excel workbook.

- The user can select an option to export without expression data.
- The user may upload a file without expression data or want to export a file without expression data.
- The selection defaults to the current expression data source.
  - If node coloring is disabled, selection defaults to the original source.
  - If all expression sheets are deselected, then the selected source automatically moves to None.

## Node

- Enable Node Coloring

Select from user-uploaded expression data, or use data from our Expression Database

## Top Dataset

Barreto\_2012\_wt

- Average Replicate Values

## Bottom Dataset

Same as Top Dataset

- Average Replicate Values

## Log Fold Change Max Value

(0.01-100):

3

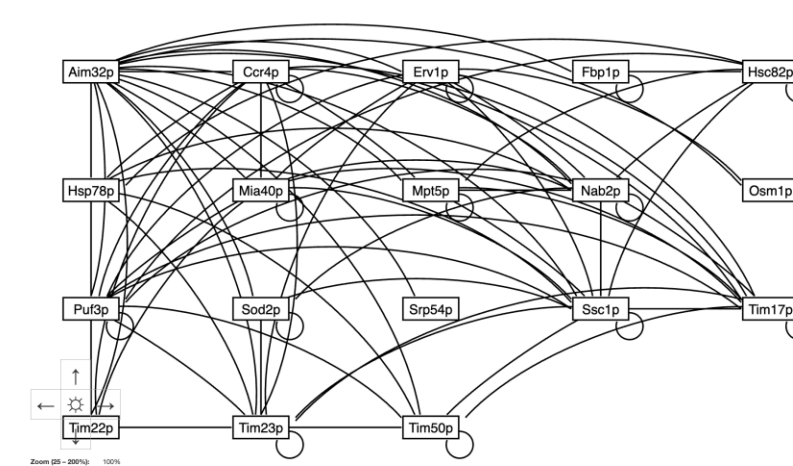
Set

-3.00 0 3.00

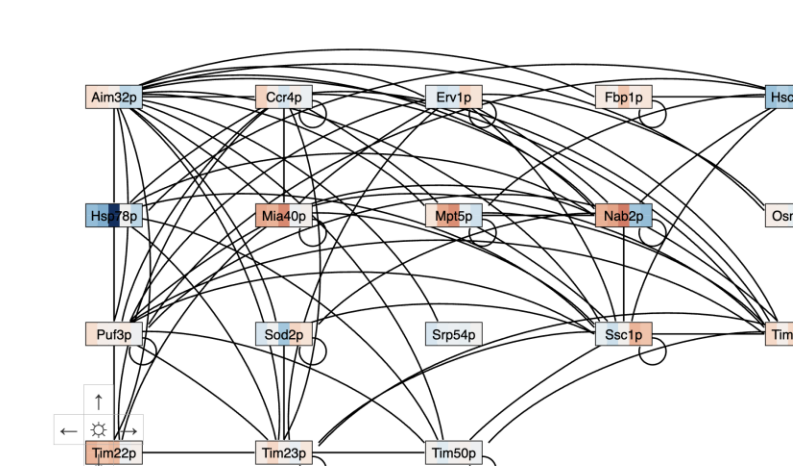
Export Workbook Cancel

Nodes in PPI networks can now be colored using mRNA expression data

- Why PPI Node Coloring?
  - Previously, node coloring was only enabled for GRNs because node coloring is a visualization of the level of mRNA expression using data provided by the user or from GRNsight's expression database.
- PPI Node Coloring Implementation
  - Now, when a PPI demo or network without expression data in the workbook is loaded, node coloring defaults to disabled.
  - When node coloring is enabled for a PPI, a warning message displays that informs users that mRNA-level expression data is being displayed on a PPI network.
  - The query on the expression database for gene names had to strip the "p" at the end of the protein name to account for the difference in yeast gene and protein nomenclature. For example, *ACE2* represents the gene and *Ace2p* represents the protein.



Demo #5: PPI with Node Coloring Disabled



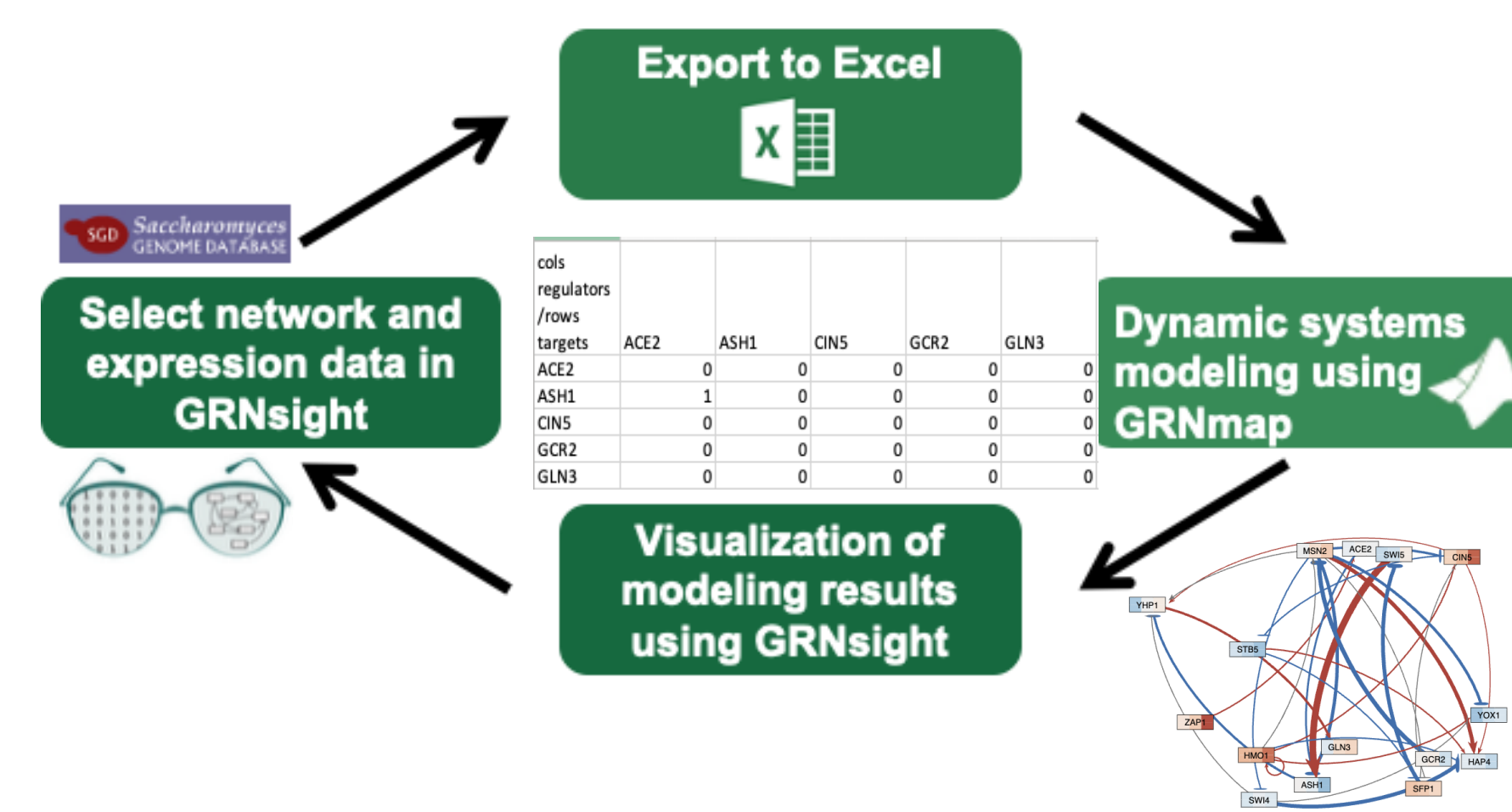
Demo #5: PPI with Node Coloring Enabled

New file naming scheme for exported files simplifies information provided to the user

- Upon export, a file is named according to type, number of genes/proteins and edges, and expression data source.
  - Eg.: GRN\_15-genes\_28-edges\_weighted\_Dahlquist\_2018.xlsx

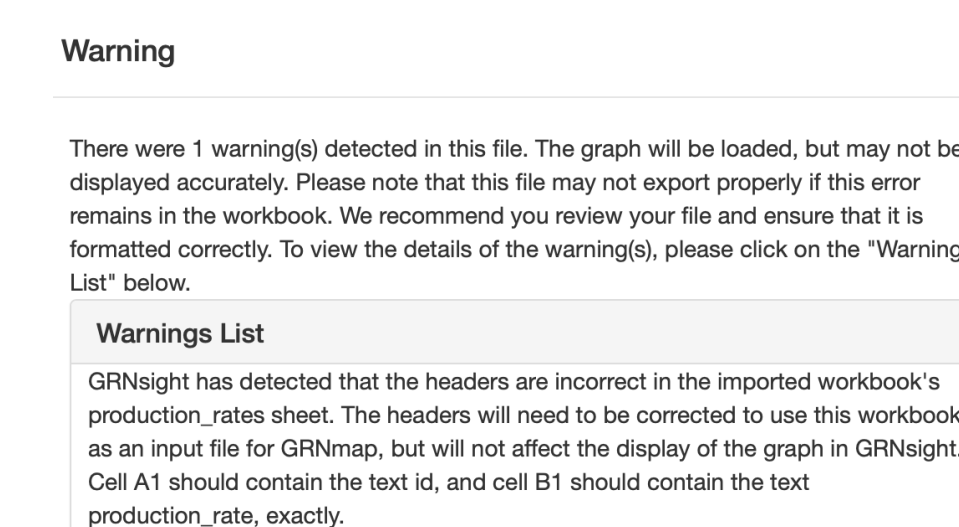
GRNsight is an integral part of a data analysis workflow for biologists who model GRNs

- GRNsight data can be exported to an Excel workbook for use with our sister MATLAB software, GRNmap.
  - GRNmap = Gene Regulatory Network Modeling, and Parameter estimation.
- Ensuring data export is free of bugs is crucial to maintaining the integrity and organization of the data while minimizing the risk of errors.
- GRNsight should also be able to import all GRNmap data and make it available for export.



New warning implementation for importing and exporting files help user catch errors

- Enhanced Error Handling
  - Comprehensive warnings have been integrated for both file import and export processes to provide users with clear, actionable feedback.



## Future Directions

- Prioritize bug resolution to enhance platform stability and reliability.
- Improve documentation for clearer and more comprehensive user guidance.
- Migrating to React to modernize the platform and improve performance.

## Availability

- GRNsight is free and open to all users at <http://dondi.github.io/GRNsight/>, and there is no login requirement.
- GRNsight code is available under the open source BSD license at our GitHub repository at <https://github.com/dondi/GRNsight>.

## Acknowledgments

We would like to thank Nikki C. Chun, Ngoc Kim Ngan Tran, Cecilia J. Zaragoza and the previous developers on the GRNsight project.

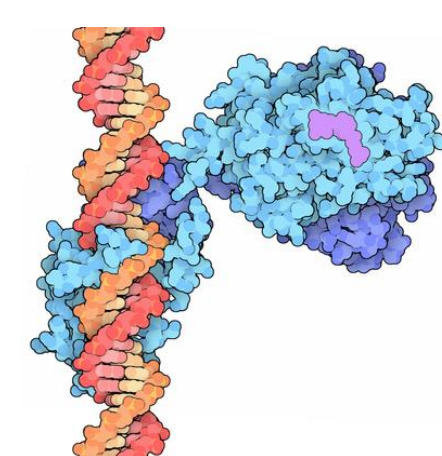
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GRNsight can also be used to visualize protein-protein physical interaction (PPI) networks

- A gene regulatory network consists of genes, transcription factors, and the regulatory connections between them which govern the level of expression of mRNA and protein from genes.

- Nodes represent both the regulatory transcription factor protein and the gene that encodes it.
- An arrow represents a regulatory relationship either activating (arrowhead, red color) or repressing (blunt end, blue color) the expression of another regulatory transcription factor.



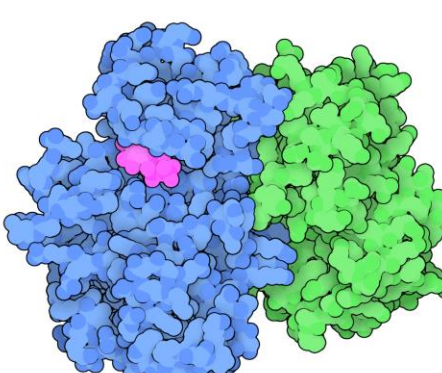
Regulatory Transcription Factor Bound to DNA  
Image from: <https://pdb101.rcsb.org/motm/155>

Cell A1 of the "network" input worksheet for a GRN:

|   | A                            | B    | C    |
|---|------------------------------|------|------|
| 1 | cols regulators/rows targets | ABF1 | ACE2 |
| 2 | ABF1                         | 0    | 0    |
| 3 | ACE2                         | 0    | 0    |

- A PPI is when two proteins bind to each other.

- Each node represents a protein, and each edge represents their binding interaction.
- Unlike GRNs, PPI graphs have undirected edges of uniform thickness and black color and previously did not display expression data on the nodes.



Protein-Protein Interaction  
Image from: <https://pdb101.rcsb.org/motm/236>

Cell A1 of the "network" input worksheet for a PPI:

|   | A                           | B      | C     |
|---|-----------------------------|--------|-------|
| 1 | cols protein1/rows protein2 | Aim32p | Ccr4p |
| 2 | Aim32p                      | 0      | 0     |
| 3 | Ccr4p                       | 1      | 1     |